

LLM-Assisted Intent-Aware Multimodal Video Retrieval with Diagnostic-Event Temporal Search

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Abstract. Interactive video retrieval is challenged by query-dependent multimodal evidence, ambiguous object descriptions, incomplete frame coverage, and temporal retrieval biased toward isolated high-scoring frames. We present a multimodal retrieval system for the AI Challenge 2026 that addresses these issues without requiring large-scale task-specific training. Our main contributions are: (1) an LLM-based retrieval planner that adaptively routes each query between semantic retrieval and textual evidence from captions, ASR, and OCR; (2) evidence-oriented reasoning that decomposes ambiguous descriptions into retrievable entities, actions, attributes, and lexical cues; and (3) Diagnostic-Event-Video-First (DEV), a coarse-to-fine global-to-local strategy that improves video coverage through adaptive sampling and performs annotation-aware temporal refinement to recover ordered event sequences rather than isolated frames. Experiments show that DEV outperforms the compared temporal retrieval methods from Recall@10 onward. In the official AIC 2026 preliminary evaluation, our system achieved 88.63% of the maximum available score and qualified as a finalist for the AIC 2026 final round.

Keywords: Event retrieval · Temporal Search · LLM-Assisted Intent-Aware Retrieval · Query-Adaptive Retrieval

1 Introduction

Interactive video retrieval requires locating relevant moments from large video collections using heterogeneous evidence, including visual content, speech, scene text, and temporal context. This problem is particularly challenging in benchmarks such as the Video Browser Showdown, Lifelog Search Challenge, and Ho Chi Minh City AI Challenge 2026 (AIC 2026)³, where queries may involve complex scenes, named entities, ambiguous objects, or temporally ordered events [4, 23, 24]. Recent systems combine vision-language embeddings

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with OCR, ASR, captions, and temporal retrieval [14, 15]; however, fixed retrieval strategies remain limited when the relevance of these evidence sources varies across queries.

First, semantic and lexical evidence are query dependent: visual descriptions favor vision-language retrieval, whereas names, dialogue, numbers, and scene text are better captured by captions, ASR, or OCR. Fixed fusion weights may therefore introduce unnecessary noise, particularly without large-scale task-specific training data. Second, challenge queries often describe objects indirectly through attributes, actions, or context rather than explicit names, making their observable evidence difficult to capture through direct query embedding. Third, sparse keyframe extraction may omit short but critical events. In temporal retrieval, direct top- K frame search can further concentrate results around a few high-scoring frames, reducing the coverage of videos containing the complete event sequence.

To address these issues, we develop a multimodal video retrieval system for AIC 2026 that combines *LLM-assisted retrieval planning* with coarse-to-fine temporal search. The planner interprets each query, transforms ambiguous descriptions into retrievable visual and textual evidence, generates complementary semantic and lexical views, and assigns query-dependent weights to visual, caption, ASR, and OCR retrieval. For temporal queries, it further decomposes the description into ordered events and estimates their importance and diagnostic value. We then introduce *Diagnostic-Event-Video-First* (DEV) retrieval, which uses global event evidence to identify promising videos before performing video-local temporal refinement under chronological and temporal-gap constraints.

Our main contributions are threefold. First, we introduce a *query-adaptive multimodal retrieval* framework with an LLM-based planner that routes queries between semantic and textual retrieval and adaptively weights visual, caption, ASR, and OCR evidence without task-specific model training. Second, we propose *evidence-oriented query reasoning*, in which the planner decomposes ambiguous or indirectly described concepts into observable entities, actions, attributes, and lexical cues, thereby generating complementary views for vector and metadata retrieval. Third, we introduce *Diagnostic-Event-Video-First (DEV)*, a coarse-to-fine global-to-local retrieval strategy that uses diagnostic events to improve candidate-video coverage before event-level temporal refinement, reducing the bias of direct top- K frame retrieval toward isolated high-scoring frames.

Experimental results show that DEV outperforms the compared temporal retrieval strategies from Recall@10 onward, indicating improved preservation of relevant videos at practical retrieval depths. In the official AIC 2026 preliminary evaluation, our complete system achieved 88.63% of the maximum available score and qualified as a finalist for the AIC 2026 final round.

2 Related Work

Recent interactive video retrieval systems increasingly combine visual embeddings, textual metadata, and temporal context to support complex search scenarios. Athena-Vision integrates Perception Encoder features with OCR, ASR, object detection, LLM-based query expansion, and Adaptive Temporal Search for multi-event retrieval [15]. Its strength lies in diversifying query perspectives while exploiting multiple evidence sources. Vortex similarly combines CLIP and SigLIP2 through reciprocal rank fusion, together with OCR, captions, speech transcripts, relevance feedback, and temporal reranking [14]. VideoEase also adopts multimodal fusion using CLIP, BLIP2, and OpenCLIP, and supports temporal search between two scenes [25]. These systems show the effectiveness of multimodal evidence fusion, but modality importance is generally not explicitly determined from the semantic structure of each query, and multi-event temporal alignment remains only partially addressed.

Other approaches focus on richer event-level and contextual representations. U-CESE retrieves coherent clips using lightweight keyframe extraction, context-aware caption propagation, and a unified clipping strategy [13]. MEMORIA combines dense annotations, CLIP retrieval, event segmentation, and LLM-assisted query analysis to improve lifelog search [8], while LifeSearch incorporates multimodal metadata and structured spatial-temporal information [20]. SnapSeek 3.0 further introduces scene-graph-based retrieval and temporal consistency mechanisms [10]. T-EAGLE represents sequences of frames as textual action narratives, enabling retrieval at the event rather than isolated-frame level [16]. U-Cker complements these methods through interfaces that support interactive exploration of temporally related lifelog moments [27]. Although these approaches improve contextual and temporal retrieval, they mainly depend on predefined temporal modes, pre-generated event representations, or temporal grouping after retrieval.

Our system builds on these advances by connecting query understanding directly to evidence selection and temporal sequence construction. An LLM generates semantic rewrites, lexical queries, event decompositions, and modality-specific evidence weights before retrieval. Diagnostic-Event-Video-First search then uses event importance and preliminary retrieval evidence to identify promising videos, after which bounded beam search explicitly aligns candidates under chronological and temporal-gap constraints. The main distinction is therefore not any individual component, but the integration of query-dependent multimodal weighting, diagnostic-event-guided retrieval, and explicit multi-event temporal alignment within a unified pipeline.

3 Architecture

Figure ?? illustrates the proposed multimodal retrieval pipeline. Given a query, the LLM planner generates semantic and lexical views, assigns modality-specific weights, and, for temporal queries, decomposes the description into ordered events with importance and diagnostic scores. Visual and metadata evidence are

retrieved in parallel and fused into hybrid frame candidates, with multiple visual branches combined by max-view aggregation and reciprocal rank fusion. For temporal queries, DEV first uses diagnostic-event evidence to select promising videos, then performs video-local retrieval with temporal redundancy reduction, chronological and gap constraints, and sequence scoring to produce the final top- K results.

3.1 Data Processing

Figure ?? illustrates our data processing pipeline. Given the raw videos, we first employ AutoShot [29] to identify shot boundaries. For each detected shot, three representative frames are sampled from the beginning, middle, and end portions. Compared with using a single representative frame, this strategy provides broader temporal coverage and reduces the risk of missing transient but retrieval-relevant visual information. The extracted keyframes, together with the original videos, are stored in Google Cloud Storage [9] for efficient media access and visualization in the retrieval interface.

Visual representation. Each keyframe is independently encoded using OpenCLIP ViT-H/14-quickgelu with DFN-5B weights [7,11] and SigLIP 2 SO400M/16 [26]. The DFN-5B checkpoint provides a strong CLIP-based image-text representation learned from carefully filtered large-scale image-text data, whereas SigLIP 2 improves semantic image-text retrieval and provides stronger multilingual and localized visual representations. Using these complementary embedding spaces increases the robustness of subsequent semantic retrieval. To obtain event-level representations, temporally adjacent keyframes whose embedding similarity exceeds a threshold T are grouped into the same event. Given an event containing N keyframes with embeddings $\{\mathbf{v}_i\}_{i=1}^N$, its representation is computed as

$$\mathbf{v}_e = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i. \quad (1)$$

This temporal aggregation reduces redundant frame-level representations while preserving the dominant semantics of each event for event-level retrieval.

Visual semantic enrichment. We further enrich each keyframe with complementary textual and object-level information. For optical character recognition (OCR), PaddleOCR [5] is employed for text detection, followed by VietOCR [19] for Vietnamese text recognition. The recognized text is subsequently normalized using the vietnamese-correction-v2 model [2] to mitigate spelling and recognition errors. Object information is extracted using the COCO-pretrained YOLO26x detector [12], which recognizes 80 common object categories. We adopt the largest YOLO26 detection variant to favor detection accuracy during offline pre-processing. In parallel, Qwen3-VL-4B-Instruct [1] generates keyframe captions describing visible entities, attributes, spatial relations, and interactions, while also providing fine-grained object-level semantic information complementary to

YOLO26 detections. This enriches the representation beyond the fixed set of object categories covered by conventional object detection.

Speech and video-level semantics. For the audio modality, speech segments are first identified using voice activity detection (VAD) and subsequently transcribed with Whisper Large V3 Turbo [18, 22]. VAD suppresses non-speech regions and provides finer speech segmentation, which is particularly useful for retaining short utterances. The resulting transcripts are post-processed by GPT-4o mini [17] for spelling normalization and English translation. In addition to local frame-level descriptions, we derive a global video-level semantic summary by providing the complete transcript of each video to GPT-4o mini and prompting it to extract its principal content. We refer to this transcript-derived summary as the *video caption*, which complements frame-level visual captions with global conversational context.

Indexing and storage. Finally, the resulting multimodal representations are organized according to their retrieval characteristics. Dense keyframe and embeddings are indexed in Milvus [28] for vector similarity search, while textual metadata, including OCR outputs, captions, object labels, transcripts, and video-level summaries, are indexed in Elasticsearch [6] and maintained together with structured metadata in PostgreSQL [21]. This separation enables the retrieval system to efficiently combine visual similarity, textual matching, and structured filtering within a unified multimedia search pipeline.

3.2 Multimodal Retrieval

Semantic Vector Retrieval We perform semantic retrieval by matching each query against the indexed keyframe representations described in Section 3.1. All query and frame embeddings are ℓ_2 -normalized before cosine-similarity retrieval.

Given the original query and its LLM-generated semantic views $\{q_j\}_{j=1}^J$, each view is independently used to retrieve its top- k keyframes. For each frame, we retain the strongest evidence across all query views:

$$s_m(f) = \max_{\substack{1 \leq j \leq J \\ f \in \mathcal{N}_k^{(m)}(q_j)}} \left[\cos \left(E_{\text{text}}^{(m)}(q_j), \mathbf{v}_f^{(m)} \right) \right]_+, \quad (2)$$

where $\mathcal{N}_k^{(m)}(q_j)$ denotes the top- k results of query view q_j under embedding space m .

When multiple embedding spaces are enabled, we combine their ranked results using weighted reciprocal rank fusion (RRF) [3]:

$$s_{\text{vec}}(f) = \sum_{m \in \mathcal{M}} \frac{\beta_m}{K + r_m(f)}, \quad (3)$$

where $r_m(f)$ is the rank of frame f under embedding space m and β_m controls its contribution.

Metadata Retrieval Visual retrieval may be insufficient for queries involving names, numbers, spoken content, or text appearing in the video. Therefore, we complement it with a metadata retrieval branch based on Elasticsearch. For each keyframe, metadata is extracted from automatic speech recognition (ASR), optical character recognition (OCR), and image captioning.

The query planner generates lexical variants for ASR/OCR and English semantic rewrites for caption retrieval. Elasticsearch performs boosted search over these sources. For a keyframe f , we define the metadata score as

$$S_{\text{meta}}(f, q) = \max_{\substack{s \in \mathcal{S} \\ q_i \in \mathcal{Q}_s}} [b_s w_s S_{\text{ES}}(f, q_i, s)], \quad (4)$$

where $\mathcal{S} = \{\text{asr}, \text{ocr}, \text{caption}\}$, S_{ES} denotes the Elasticsearch relevance score, and b_s and w_s are the source-specific boost and adaptive weights, respectively.

We then combine metadata and visual evidence as

$$S_{\text{hyb}}(f, q) = \alpha S_{\text{vis}}(f, q) + \beta S_{\text{meta}}(f, q) + \gamma S_{\text{qual}}(f). \quad (5)$$

Since the two retrieval branches operate on different score scales, we additionally apply Reciprocal Rank Fusion (RRF):

$$S_{\text{RRF}}(f) = \frac{\alpha}{k + r_{\text{vis}}(f)} + \frac{\beta}{k + r_{\text{meta}}(f)}, \quad (6)$$

where $k = 60$. This hybrid formulation supports both visual-semantic queries and queries that rely on textual evidence such as dialogue, signs, numbers, or named entities.

3.3 LLM-Assisted Retrieval Planning

We use an LLM as a query planner to transform user descriptions into structured retrieval plans containing semantic rewrites, lexical queries, and modality weights. Semantic views are generated in English for vision-language retrieval and enriched with metadata cues such as objects, actions, and attributes to better align with OCR, ASR, and other annotations.

For KIS and QA, we generate a query-level plan. For TRAKE and temporal KIS, the query is decomposed into an ordered set of events $\mathcal{E} = \{e_1, \dots, e_M\}$, each with complementary retrieval views, an importance score, diagnostic priors, and modality-specific evidence weights.

System workflow. The planning procedure consists of four stages: temporal decomposition, structured query interpretation, evidence-aware retrieval routing, event-aware reasoning.

Stage 1: Temporal and Multi-View Decomposition. For temporal KIS and TRAKE queries, the planner decomposes the description into chronologically ordered, independently retrievable events $\mathcal{E} = (e_1, e_2, \dots, e_N)$. Each event we use LLM to generate k views with Chain-Of-Thought (CoT) rules ensuring

rich in information (subject, actions, object and evidence factors (color, scene, attributes)).

Stage 2: Structured Query Interpretation. Given an input query q and its task type $\tau \in \{\text{KIS, QA, TRAKE}\}$, the LLM converts the description into a structured retrieval plan: $\mathcal{P}_i = (q_{e_i}, \mathcal{V}_i^{\text{en}}, \mathcal{V}_i^{\text{vi}}, \mathcal{T}_i, I_i, D_i, \mathbf{w}_i^{\text{modality}}, \mathbf{w}_i^{\text{metadata}})$, where q_{e_i} is the standalone event query, $\mathcal{V}_i^{\text{en}}$ denotes English semantic views, $\mathcal{V}_i^{\text{vi}}$ denotes Vietnamese semantic views, and $\mathcal{T}_i$ contains lexical queries used by text-based retrieval. I_i is the event *importance* measures the contribution of an event to the requested final result, whereas D_i is its *diagnostic prior* which estimates how useful the event is for identifying the correct video during the coarse retrieval stage. Set of modality weights $\mathbf{w}_i^{\text{modality}}$ for visual and text retrieval scores and set of metadata weights $\mathbf{w}_i^{\text{metadata}}$ for OCR, Caption and ASR retrieval scores.

Stage 3: Evidence-Aware Retrieval Routing. The planner distinguishes between visual and textual evidence rather than applying a fixed multimodal weighting to every query. At both the query level and the event level, it predicts $\mathbf{w}_i = (w_i^{\text{vis}}, w_i^{\text{text}})$, where w_i^{vis} controls visual-semantic retrieval and w_i^{text} controls metadata-based retrieval ($w_i^{\text{vis}} + w_i^{\text{text}} = 1$). Textual evidence is further decomposed into ASR, caption, and OCR sources: $\mathbf{w}_i^{\text{metadata}} = (w_i^{\text{ASR}}, w_i^{\text{CAP}}, w_i^{\text{OCR}})$, where $w_i^{\text{ASR}} + w_i^{\text{CAP}} + w_i^{\text{OCR}} = 1$.

Stage 4: Event-Aware Reasoning. Temporal structure is represented explicitly by adjacent event edges $g_i = (e_i, e_{i+1}, r_i, c_i)$, where r_i denotes the temporal relation and $c_i \in \{c_{\text{short}}, c_{\text{medium}}, c_{\text{long}}, c_{\text{unknown}}\}$ represents a semantic gap class to get information about relative temporal constraints without hallucinating exact time durations. The plan additionally distinguishes between a complete video sequence and a specific target frame through temporal intent, anchor policy.

3.4 Temporal Search

To improve candidate-video coverage in the top- K results, we propose **Diagnostic-Event Video-First Search (DEV)**, a coarse-to-fine temporal retrieval strategy for temporal KIS and TRAKE queries. DEV first determines which planned events are most useful for identifying the source video, selects a diverse set of candidate videos from global evidence, and then performs a second event-level retrieval pass restricted to those videos.

For each event e_i , the planner provides an importance value I_i , a diagnostic prior D_i , semantic and lexical query views, and temporal edges to adjacent events. We denote retrieval with candidate budget b and optional allowed-video set $\mathcal{V}$ by $R(\mathcal{P}_i; b, \mathcal{V})$. Global retrieval uses $\mathcal{V}_{\text{all}}$, whereas local retrieval uses the selected video set $\mathcal{V}^*$.

Stage 1: Global event probing. For every planned event, DEV retrieves and calibrates a bounded global candidate set $C_i^g = R(\mathcal{P}_i; b_g, \mathcal{V}_{\text{all}})$. Calibration combines rank-based and multimodal retrieval evidence. The global pass is used only to estimate video-level evidence; it does not directly produce the final temporal sequences.

Algorithm 1 Diagnostic-Event Video-First Search (DEV)

```

1: function DEV( $\{\mathcal{P}_i\}_{i=1}^M, \{I_i, D_i\}_{i=1}^M, G, K, \mathcal{H}$ )
2:   for  $i = 1$  to  $M$  do
3:      $C_i^g \leftarrow \text{CALIBRATE}(R(\mathcal{P}_i; b_g, \mathcal{V}_{\text{all}}))$ 
4:      $\rho_i \leftarrow \text{DIAGNOSTICSCORE}(C_i^g, D_i, \mathcal{H})$ 
5:   end for
6:    $\mathcal{D} \leftarrow \text{SELECTDIAGNOSTICEVENTS}(\{\rho_i\}, \mathcal{H})$ 
7:    $\mathcal{V}^* \leftarrow \text{SELECTDIVERSEVIDEOS}(\{C_i^g\}, \mathcal{D}, \{I_i\}, \mathcal{H})$ 
8:   for  $i = 1$  to  $M$  do
9:      $C_i^l \leftarrow \text{LOCALRETRIEVEANDNMS}(\mathcal{P}_i, b_l, \mathcal{V}^*)$ 
10:  end for
11:   $\Pi \leftarrow \text{RECOVERANCHOREDSEQUENCES}(\mathcal{V}^*, \{C_i^l\}, \mathcal{D}, G, \mathcal{H})$ 
12:   $\Pi \leftarrow \text{SCOREANDDIVERSIFY}(\Pi, \{I_i\}, \mathcal{D}, \mathcal{H})$ 
13:  return TOPK( $\Pi, K$ )
14: end function

```

Stage 2: Diagnostic-event and candidate-video selection. A lightweight probe estimates the discriminativeness of each event. Given N_i candidates and their distinct-video set $\mathcal{U}_i$, the concentration signal is $u_i = 1 - |\mathcal{U}_i|/N_i$. Together with the score margin m_i and top confidence c_i^{conf} , it forms a probe score $\phi_i = h_u u_i + h_m m_i + h_c c_i^{\text{conf}}$. DEV combines this retrieval-time evidence with the planner prior, $\rho_i = h_D D_i + h_\phi \phi_i$, to select the configured diagnostic events. After temporal non-maximum suppression within each event–video group, videos are ranked using diagnostic support, importance-weighted cross-event evidence, and weighted event coverage. Event-wise reservation promotes video diversity, yielding $\mathcal{V}^*$.

Stage 3: Video-local retrieval and temporal recovery. DEV performs a new, video-restricted retrieval pass $C_i^l = R(\mathcal{P}_i; b_l, \mathcal{V}^*)$ for every event; it is not a post-filter of global results. Local candidates are calibrated and temporally suppressed. For each diagnostic anchor, DEV greedily expands to the best chronologically compatible events on both sides within the same video. When timestamps are available, a transition requires $t(f_{i+1}) > t(f_i)$ and obeys the edge-specific upper-gap constraint; otherwise, DEV requires only increasing frame indices. If the diagnostic event is absent, the highest-importance available event is used as a fallback anchor.

Stage 4: Sequence scoring. Each recovered sequence π is scored using calibrated event evidence, importance-weighted coverage, diagnostic support, and penalties for missing events and excessive temporal gaps. Sequence-level non-maximum suppression and a per-video cap remove near-duplicate chains before the final top- K results are returned.

The global stage is $O(MP)$ for M events and P candidates per event, while the bounded local recovery is $O(VMb^2)$ for V selected videos and at most b retained candidates per video and event; remote retrieval costs dominate both stages in practice.

Table 1. Recall@k and latency comparison for temporal search strategies.

Method	R@1	R@5	R@10	R@20	R@50	R@100	Latency (ms)
ATS	41.03	64.10	67.95	73.08	78.21	80.77	13320.0
Vortex	37.18	61.54	69.23	75.64	82.05	88.46	10832.5
DEV	34.62	60.26	76.92	82.05	85.90	93.59	37410.0

3.5 User Interface

Our user interface is designed to support efficient human-in-the-loop video retrieval, as illustrated in Fig. ?? . The interface is organized into four main regions. **(A) Task Navigation** provides access to KIS, QA, TRAKE, and video retrieval modes. **(B) Retrieval Workspace** presents ranked keyframes together with their retrieval scores and allows users to inspect and select candidate frames. **(C) Query Panel** contains the textual query input and search controls, enabling users to formulate or refine the retrieval request. **(D) Verification and Submission** displays selected frames and their temporal context, enabling final verification before exporting the retrieval results.

4 Experiments

Research question. The experiments address three research questions. RQ1 investigates the trade-offs between retrieval effectiveness and latency of video-first retrieval with local temporal refinement versus direct temporal frame retrieval. RQ2 examines how semantic embedding model choice and the number of LLM-generated query perspectives affect retrieval effectiveness and latency. RQ3 assesses the sensitivity of retrieval performance to individual components of the LLM retrieval-planner prompt.

4.1 Temporal search algorithms (RQ1)

Experimental setup We construct an evaluation set of 78 AIC 2026 ground-truth queries, comprising 58 temporal KIS queries and 20 QA queries.⁴. We evaluated with Recall@k for $k \in \{1, 5, 10, 20, 50, 100\}$ and median latency. We compare three temporal retrieval methods: ATS, Vortex, and DEV. DEV performs global retrieval and video selection before local frame refinement. All runs use the recorded fixed configuration: GPT-4o planning, SigLIP2 visual search, temporal mode enabled, top-100 retrieval.

Result Table 1 shows that ATS performs best at early ranks, achieving the highest R@1 and R@5. In contrast, DEV obtains the best recall from R@10 onward, reaching 93.59% at R@100, compared with 88.46% for Vortex and 80.77%

⁴ Recall@k measures whether the result is retrieved within the first k candidates.

Table 2. Visual encoder choice and multi-perspective query expansion over 58 queries.

Encoder	Mode	n	MRR $\uparrow$	R@10 $\uparrow$	R@100 $\uparrow$	Lat. (ms) $\downarrow$
OpenCLIP	Full	1	0.0382	0.0862	0.1034	150.1
	View.	3	0.0067	0.0172	0.0690	207.7
	View.	5	0.0062	0.0172	0.0517	229.7
	View.	7	0.0062	0.0172	0.0517	247.6
SigLIP2	Full	1	0.0617	0.1034	0.1379	158.3
	View.	3	0.0371	0.1034	0.1379	210.7
	View.	5	0.0371	0.1034	0.1379	230.8
	View.	7	0.0369	0.0862	0.1379	249.7

for ATS. This indicates that DEV preserves more relevant candidates at larger retrieval depths.

However, this improvement comes with higher computational cost. DEV has a median latency of 37,410 ms, compared with 13,320 ms for ATS and 10,832.5 ms for Vortex, showing a trade-off between retrieval recall and runtime efficiency.

4.2 Semantic embedding models (RQ2)

Experimental setup We evaluate the effect of the visual encoder and LLM-generated multi-view query expansion on semantic retrieval. The ablation is conducted on 58 queries using the two visual encoders supported by our system, OpenCLIP and SigLIP2. For each encoder, we compare the original full query with perspective-based retrieval using $n \in \{3, 5, 7\}$ semantic views.⁵ We additionally report mean retrieval latency to quantify the computational cost of increasing the number of perspectives.

Result As shown in Table 2, full-query retrieval outperforms multi-perspective expansion for OpenCLIP, while also achieving lower latency. SigLIP2 is more robust: three or five perspectives preserve R@10 and R@100, although MRR decreases. Overall, SigLIP2 with the full query achieves the strongest retrieval effectiveness, whereas OpenCLIP with the full query provides the lowest latency. Increasing the number of perspectives beyond three yields no additional recall gains and consistently increases latency.

4.3 Prompt components for LLM Retrieval Planner (RQ3)

We keep the retrieval stack fixed and perform leave-one-component-out ablation on the planner prompt. The *Full Prompt* contains 14 components: C1 meaning preservation, C2 clause classification, C3 global weights, C4 text routing, C5

⁵ Retrieval effectiveness is measured using mean reciprocal rank (MRR), Recall@10 (R@10), and Recall@100 (R@100).

Table 3. Leave-one-component-out planner-prompt ablation. MeanR is in percent; Δ and 95% CIs are percentage points relative to Full.

ID	MeanR	Δ	95% CI	p	ID	MeanR	Δ	95% CI	p
C1	62.31	+0.54	[-2.38, 3.54]	.755	C8	63.27	+1.50	[-2.04, 5.10]	.449
C2	63.95	+2.18	[-1.70, 6.12]	.292	C9	60.75	-1.02	[-5.10, 2.79]	.646
C3	63.74	+1.97	[-1.36, 5.37]	.276	C10	63.74	+1.97	[-1.77, 6.05]	.344
C4	63.33	+1.56	[-2.38, 5.65]	.473	C11	64.42	+2.65	[-0.75, 6.26]	.160
C5	62.72	+0.95	[-2.38, 4.35]	.611	C12	63.88	+2.11	[-2.11, 6.19]	.339
C6	63.54	+1.77	[-3.27, 7.01]	.527	C13	63.40	+1.63	[-1.97, 5.31]	.415
C7	60.34	-1.43	[-5.03, 2.04]	.466	C14	62.99	+1.22	[-3.27, 5.78]	.626

modality preference, C6 English semantic views, C7 Vietnamese lexical views, C8 OCR preservation, C9 observable terms, C10 rewrite length, C11 root rewrite, C12 temporal-KIS planning, C13 task-specific event structure, and C14 event weighting.

We use `gpt-4o-mini` with temperature 0.05 and a 2,048-token output limit. The experiment covers 98 queries and 15 prompt conditions, each repeated three times, yielding 4,410 runs. All other retrieval settings are fixed: SigLIP2, hybrid top-100 retrieval, metadata enabled, temporal mode for KIS, and reranking and query expansion disabled.

We evaluate video-level Recall@ k and define MeanR as the average of Recall@1, @5, @20, @50, and @100. Exact-frame recall over the same cutoffs is used as a secondary metric. We report Δ = ablation – Full, with 95% confidence intervals from 10,000 paired bootstrap samples and two-sided p -values from 20,000 paired sign-flip tests.

The Full Prompt achieves Recall@1, @5, @20, @50, and @100 of 35.71%, 51.70%, 70.41%, 73.81%, and 77.21%, respectively, corresponding to a MeanR of 61.77% and mean exact-frame recall of 5.71%. Table 3 summarizes all ablations.

All confidence intervals include zero and all p -values exceed .05, so no individual component shows a statistically significant effect at the 5% level. C7 and C9 are the only removals that reduce MeanR, by 1.43 and 1.02 points, respectively; C9 also reduces mean exact-frame recall by 1.84 points.

The results also reveal ranking-depth trade-offs. Removing C3 improves MeanR and Recall@5 but reduces Recall@1, while removing C12 improves MeanR and Recall@100 but decreases both Recall@1 and exact-frame recall. Thus, a positive MeanR change does not imply consistent improvement across ranking depths or that a component is unnecessary, particularly because prompt components may interact.

Because all unadjusted p -values exceed .05, multiplicity correction would not change the significance conclusion. Moreover, the benchmark contains no TRAKE samples, while the QA subset and single-reference exact-frame evaluation are limited; therefore, these results should be interpreted as prompt-sensitivity analysis rather than isolated causal effects.

Table 4. Official preliminary-round results on the AIC 2026 benchmark.

Round	Official Score	Maximum Score	Score (%)
Round 1	21.27	24	88.63%
Round 2	27.20	29	93.79%
Round 3	27.75	33	84.09%
Overall	76.22	86	88.63%

4.4 Analysis

Table 4 summarizes the official preliminary-round results of our system in the AIC 2026 Multimodal Retrieval Challenge. The system achieved scores of 21.27, 27.20, and 27.75 across the three evaluation rounds, corresponding to 88.63%, 93.79%, and 84.09% of the maximum scores, respectively. Overall, the system obtained 76.22 out of 86 points, equivalent to 88.63% of the total available score. The results demonstrate the effectiveness of the proposed multimodal and temporal retrieval pipeline across multiple evaluation rounds. The lower relative performance in Round 3 indicates that further improvements are still possible for more complex retrieval scenarios.

5 Conclusion

In this work, we introduced a multimodal video retrieval system for the AI Challenge 2026 that combines LLM-assisted retrieval planning, multimodal evidence fusion, and explicit temporal reasoning. By integrating complementary visual representations with OCR, ASR, caption, and object-level metadata, the system supports both semantic and lexical retrieval across diverse query types. We further proposed Diagnostic-Event-Video-First (DEV), a coarse-to-fine temporal retrieval strategy that first identifies promising videos and then performs event-level refinement under chronological and temporal-gap constraints.

Experimental results demonstrate the effectiveness of the proposed design. DEV achieves 93.59% R@100, showing strong recall at larger retrieval depths, while the complete system obtains 88.63% of the maximum score in the AIC 2026 preliminary evaluation. These results indicate that combining query-aware multimodal retrieval with explicit temporal sequence construction is effective for complex video retrieval scenarios.

Despite these results, challenges remain in early-rank retrieval, computational efficiency, and complex multi-event scenarios. The current one-shot LLM planner may also produce redundant or suboptimal query reformulations. In future work, we plan to investigate an agentic retrieval framework that can iteratively observe intermediate evidence, select appropriate visual, textual, and temporal retrieval tools, and adapt its strategy during retrieval. We also aim to improve early-rank precision and reduce retrieval latency while preserving temporal consistency and multimodal robustness.

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